$$= \int u^9 du - a\int u^8 du$$

$$= \frac{u^{10}}{10} - a\frac{u^9}{9} + c$$

$$\int x(a-x)^8\, dx = \frac{(a-x)^{10}}{10} - \frac{a(a-x)^9}{9} + c.$$

11.7.4 Important Results

(1) $$\int \frac{f'(x)}{f(x)} dx = \log|f(x)| + c$$

(2) $$\int f'(x)[f(x)]^n dx = \frac{[f(x)]^{n+1}}{n+1} + c, \quad n \neq -1$$

Proof

(1) Let $I = \int \frac{f'(x)}{f(x)} dx$

Putting $f(x) = u$ then $f'(x)dx = du$

Thus, $I = \int \frac{du}{u} = \log|u| + c$

Therefore, $\int \frac{f'(x)}{f(x)} dx = \log|f(x)| + c.$

(2) Let $I = \int f'(x)[f(x)]^n dx$

Putting $f(x) = u$ then $f'(x)dx = du$

Thus, $I = \int u^n du = \frac{u^{n+1}}{n+1} + c$

Therefore, $\int f'(x)[f(x)]^n dx = \frac{[f(x)]^{n+1}}{n+1} + c.$

Example 11.31

Integrate the following with respect to x.

(i) $\int \tan x\, dx$ (ii) $\int \cot x\, dx$ (iii) $\int \operatorname{cosec} x\, dx$ (iv) $\int \sec x\, dx$

Solution

(i) Let $I = \int \tan x\, dx = \int \frac{\sin x}{\cos x} dx$

Putting $\cos x = u$ then, $-\sin x\, dx = du$

Thus, $I = \int -\frac{1}{u} du = -\log|u| + c = -\log|\cos x| + c = \log|\sec x| + c.$

(ii) Let $I = \int \cot x\, dx = \int \frac{\cos x}{\sin x} dx$

Putting $\sin x = u$ then, $\cos x\, dx = du$

Thus, $I = \int \frac{1}{u} du = \log|u| + c = \log|\sin x| + c.$

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